Progressive Multimodal Reasoning via Active Retrieval

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Abstract

Multi-step multimodal reasoning tasks pose significant challenges for multimodal large language models (MLLMs), and finding effective ways to enhance their performance in such scenarios remains an unresolved issue. In this paper, we propose AR-MCTS, a universal framework designed to progressively improve the reasoning capabilities of MLLMs through Active Retrieval (AR) and Monte Carlo Tree Search (MCTS). AR-MCTS follows the MCTS algorithm and heuristically integrates an active retrieval mechanism during the expansion stage to automatically acquire high-quality step-wise reasoning annotations. Moreover, we further introduce curriculum training objectives to progressively align with a process reward model, ultimately achieving process-level multimodal reasoning verification. Experimental results across three complex multimodal reasoning benchmarks confirm the effectiveness of AR-MCTS. Further analysis demonstrates that it can optimize sampling diversity and accuracy, yielding reliable multimodal reasoning.

1 Introduction

Reasoning, as the fundamental capability of large language models (LLMs) (OpenAI, 2023a; Dubey et al., 2024) and multimodal large language models (MLLMs) (Liu et al., 2023b; Bai et al., 2023), lays the foundation for generalization across a wide range of downstream tasks (Luo et al., 2023; Yuan et al., 2023; Lu et al., 2024b). In complex reasoning scenarios, models often require multiple steps to seek a final answer, resulting in various candidate reasoning paths. Therefore, efficiently identifying the correct path that includes key problem-solving steps is essential. To achieve this, reasoning verification techniques (Lightman et al., 2024; Setlur et al., 2024a; Wang et al., 2024c) employ high-quality reward models for path selection, thereby offering a promising approach to improve the reliability of model reasoning.

To improve the trustworthiness of complex reasoning, foundational efforts such as outcome reward models (ORMs) (Cobbe et al., 2021a; Shao et al., 2024) directly verify the quality of entire reasoning trajectories. However, ORMs only provide result-oriented feedback. To further obtain finer-grained verification, process reward models (PRMs) (Lightman et al., 2024; Wang et al., 2024c; Ma et al., 2023; Wang et al., 2024a) provide intermediate rewards for the model's step-wise reasoning and employ reinforcement learning algorithms for optimization (Ouyang et al., 2022). Despite these advancements, the manual annotation of reasoning paths limits its scalability and applicability (Xia et al., 2024). In response to these challenges, recent developments in inference-time scaling (Snell et al., 2024; Sardana et al., 2024) have led to the integration of the MCTS algorithm into LLMs (Browne et al., 2012). This combination allows the model to automatically sample and evaluate potential solutions at each reasoning step, designing value functions to update node weights through back-propagation, thereby achieving automatic step-wise reasoning annotation without human effort (Liu et al., 2023c; Zhang et al., 2024b,a; Yuan et al., 2024; Chen et al., 2024a).

While MCTS-based methods have been widely used in LLMs, their application to MLLMs remains largely unexplored. The unique characteristics of multimodal scenarios necessitate specialized adaptations of MCTS to address their complexities. Let us illustrate these challenges by theoretically analyzing the limitations of existing MCTS methods. At each expansion step of MCTS, given the input x and the best reasoning path y selected after simulation, the process can be modeled as:

$$p(y \mid x) = \max_{i \in k} \underbrace{p_\theta(y \mid r_i, x)}_{\text{Simulation}} \cdot \underbrace{p_\phi(r_i \mid x)}_{\text{Expansion}}, \quad (1)$$

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where r and k represent reasoning paths and the number of sampled paths, while ϕ and θ denote the generator and the verifier, respectively. This formula indicates that both expansion and simulation phases jointly determine the success of reasoning. Most existing methods focus on optimizing the simulation process, only relying on beam search based on the model's internal knowledge for the expansion phrase (Satir and Bulut, 2021; Freitag and Al-Onaizan, 2017). This simple strategy is effective for text-only reasoning tasks, as LLMs are well-pretrained on large-scale text data. However, in multimodal reasoning tasks, MLLMs often face misalignment issues between different modality inputs (Song et al., 2023; Rasenberg et al., 2020), rendering their internal knowledge insufficient for reasoning path expansion. Notably, errors at any step may disrupt the whole reasoning chain (Miao et al., 2024; Ling et al., 2023). Consequently, developing effective strategies for reliable path expansion in MLLMs poses a significant challenge in multimodal reasoning tasks.

To address these challenges, our pivotal insight is to dynamically supply relevant knowledge during each step of the reasoning expansion in the MCTS algorithm, thereby improving the reliability of multimodal reasoning. Motivated by multimodal retrieval-augmented techniques (Tan et al., 2024; Liu et al., 2023a, 2024), we propose **AR-MCTS**, a universal framework dedicated to progressively improving the complex reasoning capabilities of MLLMs through **A**ctive **R**etrieval and **M**onte **C**arlo **T**ree **S**earch. Specifically, we design a unified retrieval module to retrieve key problem-solving insights for supporting reasoning from a hybrid-modal retrieval corpus. To further achieve reliable multimodal reasoning verification, we define the quality of each reasoning step as its potential to deduce the correct answer, enabling us to iteratively obtain step-wise annotations using the MCTS algorithm. Notably, we introduce an active retrieval strategy during MCTS expansion, innovatively replacing beam search with dynamically retrieved insights to enhance sampling diversity and reliability. Based on these process-level annotations, we progressively align a process reward model tailored for multimodal reasoning through step-wise Direct Preference Optimization (DPO) (Rafailov et al., 2023a,b) and Supervised Fine-tuning (SFT) objectives, achieving automatic reasoning verification. Experiments on three complex multimodal reasoning benchmarks validate the effectiveness of AR-MCTS across various proprietary MLLMs. In summary, our contributions are as follows:

- We theoretically model the key mechanisms of the MCTS-based method in Equation (1), revealing its core limitations in multimodal reasoning.
- We are the first to introduce the retrieval mechanism in each step of multimodal reasoning to replace traditional model self-sampling strategies, enhancing both sampling diversity and accuracy.
- We propose the AR-MCTS framework, which leverages the MCTS algorithm alongside an active retrieval strategy for improving multimodal reasoning. AR-MCTS automatically acquires high-quality step-wise reasoning annotations to progressively align a process reward model, enabling reliable automated reasoning verification.
- Experimental results on three complex multimodal reasoning benchmarks demonstrate the effectiveness of AR-MCTS across various proprietary models. Quantitative analyses reveal that AR-MCTS optimizes sampling diversity and verification accuracy, providing a promising solution for reliable multimodal reasoning.

2 Related Work

LLM and MLLM Reasoning. The reasoning capabilities of LLMs (OpenAI, 2023a; Dubey et al., 2024) and MLLMs (Liu et al., 2023b; Bai et al., 2023; Chen et al., 2023b) have gained considerable attention. Techniques like Chain-of-Thought (Wei et al., 2022), Tree-of-Thought (Yao et al., 2023), and Program-of-Thought (Gao et al., 2023; Chen et al., 2023a) enhance response coherence by guiding models to decompose problems with alignment strategies. The reflection mechanism promotes iterative response improvement (Renze and Guven, 2024). Moreover, Self-Consistency (Wang et al., 2023) shows that sampling diverse solutions and selecting the most consistent one can significantly enhance LLM reasoning. Robust reward models are also effective in optimizing response quality (Setlur et al., 2024b; Uesato et al., 2022; Wang et al., 2024a; Li et al., 2023). Several studies further enhance the reasoning capabilities of large language models by integrating a diverse set of external tools, including search engines and Python interpreters (Dong et al., 2025; Li et al., 2024b; Zelikman et al., 2022; Li et al., 2025a,b). Recently,

o1-level LLMs introduce a "slow thinking" mechanism that integrates large-scale RL, MCTS algorithm and PRM to improve the step-wise reasoning (Zhang et al., 2024b; Gao et al., 2024; Zhang et al., 2024a). Despite this progress, the reasoning potential of MLLMs remains underexplored.

Multimodal Retrieval-Augmented Generation. Recent Retrieval-Augmented Generation has shown exceptional performance across various NLP tasks by incorporating relevant information from diverse sources (Jin et al., 2024; Asai et al., 2024; Shi et al., 2024; Yoran et al., 2024; Yu et al., 2023; Jin et al., 2025; Li et al., 2024c; Dong et al., 2023; Cheng et al., 2024; Dong et al., 2024c; Tan et al., 2025; Lei et al., 2023; Luo et al., 2024a). This technique also enhance reasoning and question-answering in the multimodal domain (Lim et al., 2024; Tan et al., 2024; Liu et al., 2023a; Zhou, 2024). However, the reasoning process is largely unexplained and lacks verification mechanisms. In this paper, we propose an active retrieval strategy that retrieves multimodal information at each step to align the PRM, facilitating reliable reasoning.

3 Methodology

We propose the AR-MCTS framework to achieve step-wise reasoning verification through active retrieval and Monte Carlo tree search. As shown in Figure 2 & 3, AR-MCTS consists of two main components: 1) It introduces a unified retrieval module, including a high-quality hybrid-modal retrieval corpus (§3.2) and a multimodal retrieval module (§3.3). This module employs knowledge concept filtering to select key insights for problem-solving (§3.4). 2) It automates the acquisition of step-wise annotations for multimodal reasoning using MCTS and an active retrieval mechanism (§3.3). Then, it leverages the annotated data to progressively align the PRM in two stages (§3.6), allowing for fine-grained verification of MLLM reasoning. Below, we will first introduce the task definition (§3.1) and then we delve into the specifics of our approach.

3.1 Preliminary

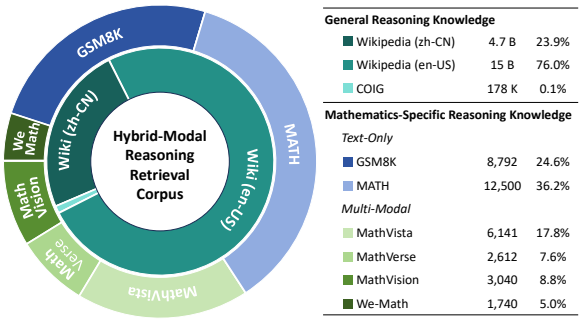
Monte Carlo Tree Search. MCTS is a widely used sampling-based search method for decision-making optimization. Its core algorithm consists of four steps: selection, expansion, evaluation, and back-propagation. By repeatedly executing these four steps, it constructs a search tree. During the selection phase, MCTS recursively selects child nodes from the root using the Upper Confidence Bound (UCB) (Srinivas et al., 2012):

$$\mathrm{UCB}(i) = w_i + C * \sqrt{2 * \ln \frac{N_i}{n_i}}, \quad (2)$$

Problem Formulation. Formally, in multimodal reasoning, given a multimodal query Q^m and corresponding retrieved problem-solving insights r from the retrieved hybrid-modal corpus D_H, we assume that the MLLM π_θ operates in an auto-regressive manner to generate a reasoning path of k steps $[y_1, \ldots, y_k]$:

$$p_\theta(\boldsymbol{y} \mid Q^m, R) = \prod_{i=1}^{k} p_\theta\left(y_i \mid Q^m, r, y_{<i}\right). \quad (3)$$

In this paper, we obtain different intermediate reasoning trajectories as the MLLM decodes to a specific termination token. Following the setup of (Tian et al., 2024), we formulate the generation process as a Markov Decision Process (MDP) (Puterman, 1990) and adopt sentence-level MCTS modeling. In reinforcement learning terminology (Schulman et al., 2017), we define the current decoded intermediate step y_i as a state s_i, corresponding to a leaf node. The process of backtracking to sample the next step is considered an action a_i. [1]



General Reasoning Knowledge		
Wikipedia (zh-CN)	4.7 B	23.9%
Wikipedia (en-US)	15 B	76.0%
COIG	178 K	0.1%
Mathematics-Specific Reasoning Knowledge		
Text-Only		
GSM8K	8,792	24.6%
MATH	12,500	36.2%
Multi-Modal		
MathVista	6,141	17.8%
MathVerse	2,612	7.6%
MathVision	3,040	8.8%
We-Math	1,740	5.0%

Figure 1: The statistics of our hybrid retrieval corpus.

3.2 Hybrid-Modal Retrieval Corpus Construction

In an ideal scenario, enhancing reasoning capabilities through retrieval resembles giving MLLMs an open-book exam. However, the multimodal reasoning field suffers from a persistent lack of high-quality reasoning retrieval corpora. To systematically establish a high-quality reasoning retrieval library, we conduct a comprehensive survey of open-source datasets, emphasizing both general

[1] Detailed definitions of MCTS are listed in the Appendix.

and mathematics-specific reasoning knowledge in multimodal contexts.

Mathematics-Specific Reasoning Knowledge. Mathematical reasoning is an essential skill of fundamental models, supported by a series of high-quality datasets. In the text-only aspect, we select the widely used math datasets, GSM8K (Cobbe et al., 2021b) and MATH (Hendrycks et al., 2021). For the multimodal domain, we adopt four meticulously cleaned math datasets: MATH-VISTA (Lu et al., 2024b), MathVerse (Zhang et al., 2024c), MathVision (Wang et al., 2024b), and WE-MATH (Qiao et al., 2024). To further prevent data leakage, we filtered out any overlapping portions with our testing benchmark using regular expressions, concatenating each sample's question q, solution process p, and answer a into a single text format, along with the corresponding image storage paths. Ultimately, we obtain 22K text-only QA pairs and 12.5K multimodal sample pairs as proprietary sources D_M from six data sources, encompassing over 20 mathematical sub-fields, with each sample containing detailed solution steps.

General Reasoning Knowledge. In the real world, general reasoning extends beyond natural subjects. To meet this broader need, we follow the conventional RAG approach (Lewis et al., 2020; Zhao et al., 2024) by utilizing Wikipedia as a web-based retrieval source, alongside the COIG (Zhang et al., 2023) large-scale question bank for general reasoning. We conduct thorough data cleaning and chunking operations, ultimately constructing this extensive dataset as our general reasoning knowledge base D_G. The statistical information of our hybrid-modal reasoning corpus $D_{\text{H}} = D_M \cup D_G$ is presented in Figure 1. [2]

3.3 Unified Multimodal Retrieval Module

Given a text-image pair from the multimodal test set $Q^m = \{x, t\}$, our goal is to retrieve the top-K multimodal relevant knowledge for each sample. Since our retrieval library encompasses hybrid-modal retrieval sources, two retrieval processes are considered to obtain the top-K pairs:

Text Retrieval. Given a text query q for multimodal sample, we use a dense retriever to retrieve k relevant documents $D_q = \{d_i\}_{i=1}^{k}$ from a text-only corpus. In this work, we employ Contriever (Izacard et al., 2022) to obtain hidden vectors for both

[2]Details of retrieval corpus processing and data contamination analysis can be found in Appendix B.4 and Table 5

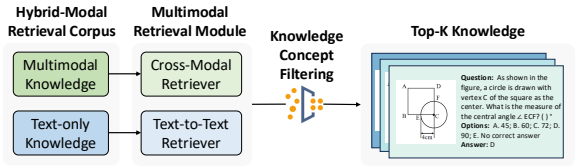


Figure 2: The pipeline of our unified multimodal retrieval module.

queries and documents. The relevance score is calculated by computing the dot-product similarity between their representations, which facilitates the retrieval of the Top-K documents D_q as follows:

$$D_q = \text{argtop}_k^{i=1,\ldots,N} \left[E_{\text{d}}(d_i)^\top \cdot E_{\text{q}}(q) \right]. \quad (4)$$

Cross-modal Retrieval. We utilize widely used contrastive vision-language models CLIP (Radford et al., 2021), which employs a dual-stream architecture featuring an image encoder $E_I(\cdot)$ and a text encoder $E_T(\cdot)$. Furthermore, we use CLIP to encode image-text pairs (x, t), obtaining the image and text vectors $E_I(x)$ and $E_T(t)$. Since the hybrid-modal retrieval corpus contains both multimodal and text-only samples, we follow previous work (Tan et al., 2024) to derive encoding vectors for the entire hybrid-modal corpus D_{H} as follows:

$$E_x(x,t) = \begin{cases} \frac{E_I(x)+E_T(t)}{2}, & \text{if } t \neq \emptyset \text{ and } x \neq \emptyset, \\ E_T(t), & \text{if } t \neq \emptyset \text{ and } x = \emptyset. \end{cases} \quad (5)$$

where $\emptyset$ denotes the empty set. For the i-th multimodal query Q^m, we encode it into a mixed vector $E_x(Q^m) = \frac{E_I(x)+E_T(t)}{2}$. We perform cross-modal retrieval between the encoding of each multimodal query and the entire retrieval database, utilizing FAISS (Johnson et al., 2021) for indexing to retrieve K samples for each query:

$$D_{\text{cross}} = \text{argtop}_k^{j=1,\ldots,N} \left[E_x(Q^m)^\top \cdot E_x(x_j, t_j) \right]. \quad (6)$$

Here, $E_x(Q^m)$ and $E_x(x_j, t_j)$ denote the embeddings of the multimodal query and the samples in the hybrid-modal corpus.

3.4 Knowledge Concept Filtering

During our deployment process, we observe that multimodal reasoning with retrieved knowledge is highly sensitive to the consistency of fine-grained knowledge concepts (*e.g.*, *algebra* knowledge can't help in solving *triangles* problems). Most high-quality visual math benchmarks provide detailed category labels (*e.g.*, *"Angles and Length"*) for

each sample, motivating us to consider knowledge concepts for fine-grained filtering. Given a multimodal query Q^m and its knowledge concept label L_{kc}, we encode the top-K retrieved hybrid-modal samples from $D_{\text{H}} = \{D_{\text{q}} \cup D_{\text{cross}}\}$ according to Equation (5) and compute the similarity with the knowledge concept's embedding $E_T(kc)$ following the pipeline in "Cross-Modal Retrieval". We strictly enforce the original retrieval similarity threshold T_r and the knowledge concept consistency threshold T_{kc}, allowing only those samples that meet both criteria to serve as key insights D_{ins} for the query Q^m:

$$D_{\text{ins}} = \{r \in D_{\text{H}} \mid \text{Sim}(r, Q^m) \geq T_r \ \& \ \text{Sim}(r, L_{\text{kc}}) \geq T_{\text{kc}}\},$$

where $\text{Sim}(x, y)$ represents the cosine similarity between the embeddings $E(x)$ and $E(y)$, $r \in D_{\text{H}}$ denotes a retrieved insights from the corpus D_{H}. Detailed information on the filtering process can be found in the Appendix.

3.5 Progressive Multimodal Reasoning Annotation

As shown in Figure 3, we will present our detailed algorithm design of AR-MCTS, which includes four core operations:

- **Selection.** During the j-th simulation of the AR-MCTS, the process begins with s_0, representing the initial state containing the multimodal input query $Q_0^m = (x_0, t_0)$ and corresponding retrieved problem-solving insights r_0. The algorithm proceeds to explore the MCTS by selecting as Equation (2) iteratively, then we formulate the multimodal query of the state s_j as $Q_j^m = \{(x_j, t_j) \mid t_j = t_0 + \sum_{i=1}^{j} y_i\}$,

- **Expansion with Active Retrieval Strategy.** Given the state s_i represented by the selected leaf node, the MCTS-based approach backtracks to the prior state, forming our multimodal input (x_i, t_i, r_i). The temperature in the traditional expansion process is empirically increased to greater than 0.6 to sample potential candidate actions for the next step (Chen et al., 2024b). Unlike them, we emphasize that the supporting knowledge required for different reasoning trajectories at each step should vary, and propose an **Active Retrieval** strategy. As shown in Figure 3, during the MCTS expansion phase at state s_i, we first concatenate the input Q_i^m with the previous reasoning steps. Then we dynamically retrieve the required candidate insights r_i for each step from the problem-solving insight library D_{ins} according to Equation (6), and replace the insight r_{i-1} from the previous step with the latest retrieved insights r_i. According to the Equation (3), the process of sampling k reasoning paths at state s_i can be modeled as follows:

$$p_\theta(y \mid x) = \prod_{i=1}^{k} p_\theta\left(\{y_i^j\}_{j=1}^k \mid Q_i^m, r_i\right). \quad (7)$$

- **Simulation.** We use the probability of deducing the correct answer based on partial solutions as a criterion for quality assessment. Following Wang et al., we apply a one-step rollout for each node obtained during expansion to ensure efficiency. We construct a value function as $V(s_i) = \frac{\sum_{j=1}^{k} \mathbb{I}(y_j = \hat{y}_i)}{k}$, where k, $\mathbb{I}$ denotes the number of sampled reasoning paths and the indicator function. If the final answer y_j equals the grounding truth $\hat{y}_i$, we set the value of the current node to 1; Otherwise, we set it to 0.

- **Back-Propagation.** For the terminal nodes reached during the rollout and the current leaf node, MCTS performs a backward update of the visit count and Q-value for each (s, a) along the route from the current node to the root, which is formulated as $N(s, a) \leftarrow N(s, a) + 1$, $Q(s, a) \leftarrow Q(s, a) + \frac{1}{N(s,a)}(V(s) - Q(s, a))$.

3.6 Curriculum Process Reward Modeling

After acquiring step-wise reasoning annotations, we draw inspiration from curriculum learning (Sun et al., 2024; Dong et al., 2024b) to design a two-phase approach for PRM. In the first stage, the model learns to distinguish the correctness of reasoning steps. Then, it learns to assign scores to each step, facilitating generalization from easy to hard.

Step-wise DPO Pre-alignment. Each round of expansion and evaluation in AR-MCTS naturally generates batches of positive and negative pairs, inspiring us to align preferences using step-level Direct Preference Optimization (DPO) as the objective. Under the state s_i (i-th reasoning step), given a multimodal query Q_i^m and a sampled reasoning path set $Y_i = \{y_j\}_{j=1}^k$, along with the corresponding value set $V_i = \{v_j\}_{j=1}^k$. We filter the paths in Y_i with value $v_j > 0.8$ as positive samples y_j^+,

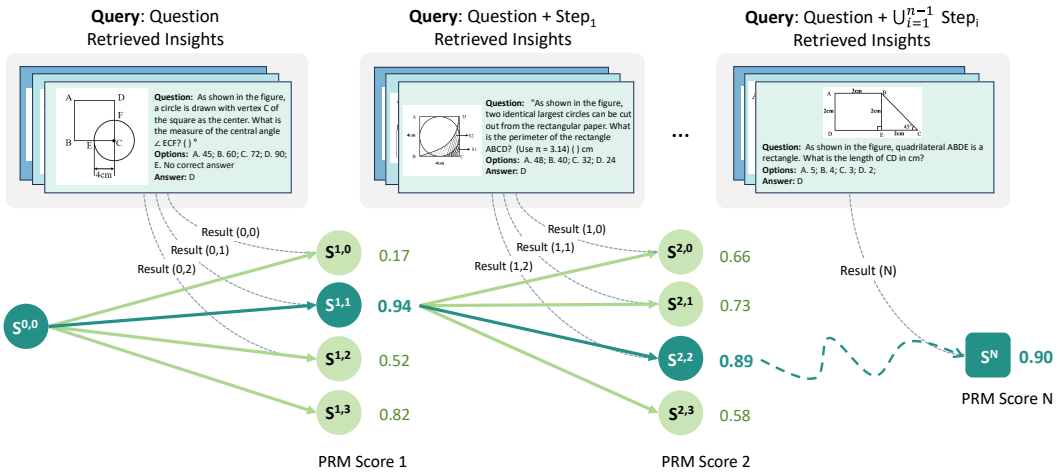


Figure 3: Our AR-MCTS: The retrieval module actively retrieves key insights at each step of the MCTS process

while those with $v_j = 0$ are regarded as negative samples y_j^-. Therefore, for each problem Q_i^m, we can obtain K pairs of step-level preference pairs $D_i^{\text{step}} = (y_j^+, y_j^-)_{j=1}^K$ and follow step-level DPO to align the reasoning discernment capability:

$$\mathcal{L}_{\text{SDPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(Q^m, y^+, y^-)\sim\mathcal{D}^{\text{step}}}[\log\sigma(\beta\log \frac{\pi_\theta(y^+|Q^m)}{\pi_\theta(y^+|Q^m)} - \beta\log\frac{\pi_{\text{ref}}(y^-|Q^m)}{\pi_{\text{ref}}(y^-|Q^m)})],$$

The reference model π_{ref} is initially set to π_θ and remains constant during training. β is a hyperparameter, and σ denotes the sigmoid function. The objective of $\mathcal{L}_{\text{SDPO}}$ is to maximize the likelihood of preferred y^+ compared to the dispreferred y^-.

Point-wise Fine-tuning. After pre-alignment, PRM has gained the initial ability to distinguish the correctness of step-wise reasoning. To further unlock its reasoning scoring capability, we apply a step-level cross-entropy objective to the pre-aligned PRM π_{DPO} as follows:

$$\mathcal{L}_{\text{PFT}} = \sum_{i=1}^{N} \left[\hat{y}_i \log_{\pi_{\text{DPO}}}(r_i) + (1-\hat{y}_i)\log_{\pi_{\text{DPO}}}(1-r_i)\right], \tag{8}$$

where $\hat{y}_i$ is the golden label $(0, 1)$ for the state s_i, r_i is the sigmoid score assigned by PRM. Finally, we progressively achieve a well-aligned PRM.

Inference. We use PRM to evaluate each step as outlined in Figure 3. Following Luo et al., we adopt point-wise soft labels and discuss PRM's hard labels in the appendix. Unlike the training data annotation, we extract the highest-scoring node from K expanded reasoning paths in each round, pruning low-quality paths. Besides, we set an early stopping criterion at the 4th round to reduce computational complexity. [3]

4 Experiments

4.1 Experimental Setup

We perform experiments on two widely used multimodal mathematical reasoning benchmarks: MATHVISTA (Lu et al., 2024b) and WE-MATH (Qiao et al., 2024). To further validate our AR-MCTS in the general reasoning domain, we perform cross-domain evaluation on the GAOKAO-MM benchmark (Zong and Qiu, 2024). For baselines, we employ AR-MCTS on strong proprietary and open-source models: (1) Closed-source MLLMs: GPT-4o (OpenAI et al., 2024), GPT-4V (OpenAI, 2023c); (2) Open-source MLLMs: LLaVA-OneVision-Qwen2-72B (Li et al., 2024a), InternVL2-8B (Chen et al., 2024d), Qwen2-VL-7B (Wang et al., 2024d), LLaMA3-LlaVA-NeXT-8B (Liu et al.). Referencing relevant works on MCTS (Wang et al., 2024c; Tian et al., 2024), we implement Self-Consistency (Wang et al., 2023), Self-Correction (He et al., 2024), and ORM (Cobbe et al., 2021a) as our core comparison strategies.

4.2 Overall Results

Table 1 illustrates the main results. Overall, AR-MCTS significantly improves visual reasoning performance across various MLLMs and reasoning verification strategies on two benchmarks, conclusively demonstrating the advantages of our approach. Below, we identify the following insights:

1) MLLMs struggle to self-correct reasoning errors. The self-correction strategy strug-

[3] Details of PRM training guideline, data collection and composition analysis are listed in Appendix B.7, B.6 & 4.6.

Table 1: Mathematical reasoning assessment on different MLLMs using MATHVISTA and WE-MATH *testmini* Sets. Following MathPUMA (Zhuang et al., 2024), we selected six reasoning-related domains for MATHVISTA. The top scores for each model are highlighted in **bold**. For more details, please refer to the Appendix B.

Model	Method	MATHVISTA						WE-MATH					
		ALL ↑	GPS ↑	MWP ↑	ALG ↑	GEO ↑	STA ↑	S3 ↑	AVG ↑	IK ↑	IG ↑	CM ↑	RM ↓
GPT-4o	Zero-shot	59.0	59.6	65.1	61.2	60.7	72.4	46.1	40.8	**31.8**	13.7	33.9	37.8
	Self-Consist.	61.8	68.3	65.1	**68.0**	68.2	74.8	53.0	45.2	29.9	12.8	38.8	32.8
	Self-Correct.	59.9	61.1	65.6	61.2	61.1	72.8	43.6	42.9	31.2	**15.2**	35.2	34.2
	ORM	61.9	68.3	66.1	**68.0**	68.2	74.8	50.3	44.3	26.5	10.9	38.9	38.0
	AR-MCTS	**62.6**	**68.6**	**66.4**	**68.0**	**68.8**	**75.3**	**56.4**	**46.8**	28.0	12.8	**40.4**	**31.8**
LLaVA-OneVision-72B	Zero-shot	64.2	**80.8**	69.4	73.3	**77.0**	66.8	40.6	24.6	42.5	14.1	17.5	59.7
	Self-Consist.	66.0	79.8	**73.1**	74.0	76.6	**67.8**	38.2	36.9	33.9	15.8	**29.0**	42.4
	Self-Correct.	58.3	78.4	68.8	70.1	74.9	56.8	30.3	14.7	**55.4**	11.8	8.7	73.3
	ORM	65.9	80.3	**73.1**	74.0	**77.0**	**67.8**	**44.2**	30.6	34.9	**18.1**	21.5	54.3
	AR-MCTS	**66.3**	79.8	**73.1**	**74.4**	76.6	**67.8**	**38.9**	**37.4**	33.7	**18.1**	28.4	**41.1**
InternVL2-8B	Zero-shot	57.3	62.5	62.4	61.2	60.7	59.1	23.6	17.4	59.8	10.1	12.4	58.9
	Self-Consist.	61.8	**77.4**	64.0	**73.0**	**72.8**	62.1	35.1	26.6	45.5	13.5	19.8	51.6
	Self-Correct.	46.8	57.7	31.2	55.9	56.1	46.2	30.3	9.8	**62.7**	8.6	5.5	80.8
	ORM	61.1	67.8	64.0	64.1	64.9	68.4	32.7	29.7	42.9	**16.0**	21.7	**47.2**
	AR-MCTS	**63.1**	62.9	**71.6**	59.9	62.6	**71.4**	**43.6**	**30.5**	37.7	14.7	**23.2**	51.2
Qwen2-VL-7B	Zero-shot	58.8	45.5	60.5	45.5	47.9	70.8	33.9	19.8	51.2	12.6	13.5	62.6
	Self-Consist.	61.2	54.8	61.8	56.2	55.2	72.1	33.9	23.6	46.9	13.7	16.8	57.5
	Self-Correct.	50.8	43.3	53.2	45.9	43.9	62.1	26.7	20.0	**54.1**	11.1	14.5	58.5
	ORM	62.3	55.5	62.7	56.9	56.5	**72.4**	34.6	26.4	42.9	11.2	20.8	54.8
	AR-MCTS	**64.1**	**63.9**	**72.6**	**60.9**	**63.6**	**72.4**	**40.6**	**28.1**	40.0	**14.3**	**21.0**	**54.2**

gles across reasoning benchmarks. Although a minor improvement is noted with GPT-4o, other weaker open-source MLLMs experience significant declines after the self-correction, particularly Qwen2VL-7B, which shows a drop of over 8% on MATHVISTA (ALL). This discovery corresponds with the findings of He et al., highlighting the instability of correction methods that rely on the self-knowledge of MLLMs, especially in small MLLMs.

2) PRM outperforms ORM in complex reasoning. Compared to ORM, AR-MCTS with PRM demonstrates a more significant performance improvement across most MLLM backbones on the S3 metrics in WE-MATH (GPT-4o: 56.4% vs 50.3%; Qwen2-VL: 40.6% vs 34.6 %). This highlights that PRM, by meticulously verifying each step of the reasoning process, achieves stronger alignment in multi-step reasoning tasks.

3) AR-MCTS better unlocks the reasoning potential of weaker MLLMs. Compared to LLaVA-OneVision-72B, Qwen2-VL-7B with AR-MCTS shows a significant improvement over the zero-shot setting on MATHVISTA (ALL: 5.3%↑) and in WE-MATH (AVG: 8.3%↑). A similar finding is observed with InternVL2-8B, indicating that the performance gains of AR-MCTS are more pronounced in smaller MLLMs. This observation further verifies the importance of integrating active retrieval in reasoning. AR-MCTS is a reliable plug-and-play framework, offering a promising solution for reasoning alignment in weaker MLLMs.

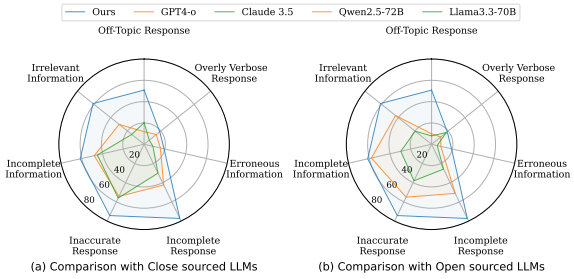


Figure 4: The results of MLLMs on GAOKAO-MM.

4.3 General Reasoning Domain Verification

To validate the effectiveness of AR-MCTS in the general multimodal reasoning field, we further evaluate the Chinese human-level reasoning benchmark, GAOKAO-MM. As shown in Figure 4, both the closed-source GPT-4o and the open-source small MLLM Qwen2-VL-7B with AR-MCTS framework demonstrate significant improvements over the backbone and self-consistency approaches, verifying the generalization of AR-MCTS across different languages and disciplines. Notably, AR-MCTS with GPT-4o achieves stable improvements in mathematics and physics (12.5% ↑ and 7.7% ↑), while also showing some gains

Table 2: Ablation study with Qwen2-7B. "*w/o* Active Retrieval" corresponds to the "vanilla PRM baseline", while "*w/o* PRM" denotes using "Beam search with retrieval". "Filtering" is the "knowledge concept filtering".

Models	MATHVISTA (ALL)	WE-MATH (S3)	GAOKAO -MM(ALL)
AR-MCTS	64.1	40.6	37.4
w/o PRM	61.0 (-3.1)	37.7 (-2.9)	33.2 (-4.2)
w/o Filtering	62.8 (-1.3)	39.5 (-1.1)	34.5 (-2.9)
w/o Active Retrieval	61.9 (-2.2)	38.7 (-1.9)	33.4 (-4.0)

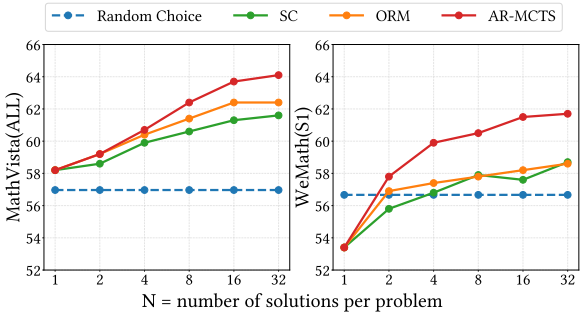


Figure 5: Scaling analysis on inference samplings. *Random Choice* denotes randomly sampling from 1 to 32.

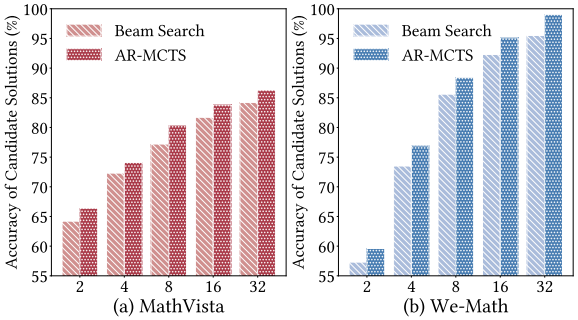


Figure 6: The comparison of solution sampling.

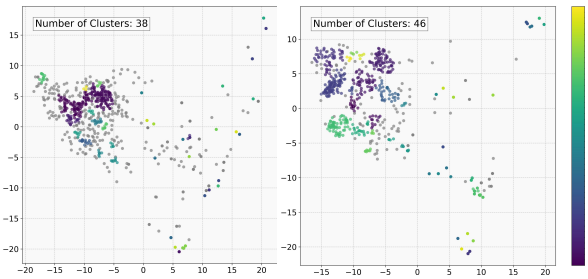


Figure 7: The visualization of the sampled solutions.

in the humanities (e.g. history 20%↑). This emphasizes that AR-MCTS with PRM not only improves the complex reasoning abilities of MLLMs, but also effectively mitigates the knowledge gaps of MLLMs in the humanities through its retrieval mechanism.

4.4 Quantitative Analysis

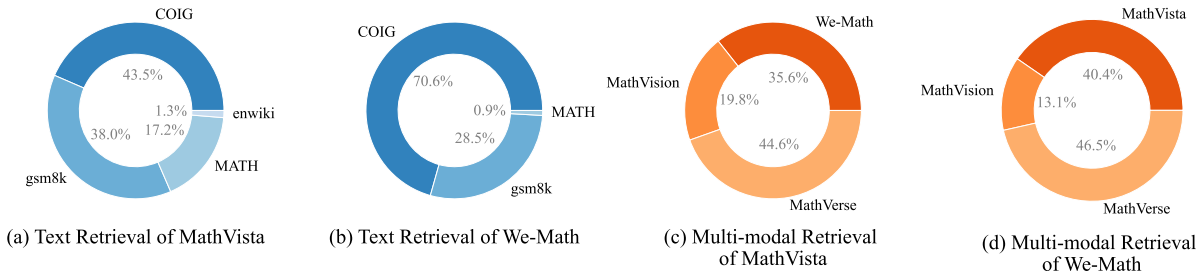
Ablation Study. To explore the effects of components in AR-MCTS, we conduct an ablation study in Table 2. The term "w/o" indicates versions without specific components. Our key findings are: 1) Removing any component from AR-MCTS results in performance decline, highlighting the necessity of all component designs. 2) Removing the PRM or active retrieval causes a significant performance drop (MATHVISTA: 3.1% and 2.2%), highlighting that both step-wise verification and active retrieval effectively enhance MLLM's reasoning capabilities. 3) Notably, knowledge concept filtering achieves stable performance gains, indicating that it effectively reduces noise in retrieved knowledge and highlights the importance of consistency between the retrieved knowledge and the problem during reasoning.

Scaling Analysis on Inference Sampling. As shown in Figure 5, we perform a scaling analysis to evaluate the performance of various strategies across two benchmarks with sampled paths ranging from 1 to 32. Our main conclusions are listed: 1) When the number of sampled solutions exceeds a certain threshold (16), self-consistency (SC) exhibits performance fluctuations in WE-MATH. 2) AC-MCTS consistently outperforms ORM and SC, with this superiority becoming more pronounced as N increases. We attribute this advantage to our automated process labeling, which offers high scalability and low annotation costs while providing more reliable feedback for path verification.

4.5 Does AR-MCTS Improve Sampling?

In this section, we explore whether AR-MCTS can efficiently improve the quality of the candidate solution sampling from the following two perspectives:

Accuracy Analysis. To validate that AR-MCTS can efficiently improve the solution sampling accuracy in multimodal reasoning, we analyze the "Correctness of questions" of Qwen2-VL during the sampling process in the two benchmarks. The accuracy can be formulated as $P_Q^c = \frac{N_Q^c}{N_Q}$, where N_Q^c denotes the number of questions containing at least one correct candidate solution, while N_Q denotes the number of questions. As shown in Figure 6, AR-MCTS demonstrates consistent gains in both benchmarks compared to the traditional beam search sampling. As the number of candidate solutions increases, the answer accuracy P_Q^c exhibits a positive correlation. This finding confirms that AR-MCTS is a scalability framework that ef-



(a) Text Retrieval of MathVista (b) Text Retrieval of We-Math (c) Multi-modal Retrieval of MathVista (d) Multi-modal Retrieval of We-Math

Figure 8: The composition analysis on retrieval corpus of WE-MATH and MATHVISTA.

ficiently improves the reliability of the sampling space in multimodal reasoning, thereby addressing the inherent challenges of MCTS-based methods.

Diversity Analysis. To investigate whether AR-MCTS enhances the diversity of sampled solutions, we sample 250 problems from MATHVISTA and use AR-MCTS to sample 4 candidate solutions for each, resulting in 1,000 samples. We use BGE-M3 (Chen et al., 2024c) as our semantic embedding model, apply PCA for dimensionality reduction, and use DBSCAN (Ester et al., 1996) clustering to visualize the solution representations.

Figure 7 shows the visualization between the beam search (left) and AR-MCTS (right). The representations of candidate solutions sampled by beam search tend to collapse into a small area with several noise points (in gray), reflecting that the beam search may lead to redundancy in sampling. Under the same setting, AC-MCTS clusters more centroids for the same problem set (38 vs.46) and exhibits a more dispersed distribution. This verifies that AR-MCTS alleviates the issue of limited diversity in solutions sampling, efficiently covering the problem-solving space and providing strong prior conditions for the simulation process of MCTS.

4.6 The Composition Analysis of Retrieval Knowledge

To gain deeper insights into which knowledge sources provide the greatest benefits to our multimodal reasoning test set, we conduct a comprehensive ranking of the hybrid-modal knowledge retrieved from samples of MATHVISTA and WE-MATH based on similarity, selecting the Top-50 relevant knowledge samples and visualizing their respective sources. As shown in Figure 8, both MATHVISTA and WE-MATH exhibit significant diversity in their retrieved knowledge, whether from text-only or multimodal sources. This highlights the motivation for constructing our hybrid-modal retrieval library from diverse, high-quality reasoning datasets. It also confirms that the insights needed for problem-solving do not necessarily originate from the same type of data source but should be enhanced through diverse reasoning knowledge. Our hybrid-modal reasoning retrieval library effectively addresses this need.

5 Conclusion

In this paper, we introduce AR-MCTS, a universal framework designed to enhance complex multimodal reasoning capabilities. AR-MCTS integrates the MCTS algorithm with an active retrieval strategy to automatically acquire high-quality stepwise reasoning annotations, progressively aligning a PRM for process-level multimodal reasoning verification. Experiments confirm the effectiveness of AR-MCTS across various MLLMs and benchmarks, demonstrating its ability to optimize sampling diversity and verification accuracy, thus providing a promising solution for reliable reasoning.

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A More Details about AR-MCTS

A.1 The Algorithm Workflow of AR-MCTS

In this section, we will explore the overall workflow of AR-MCTS, highlighting its key components and steps involved in the retrieval and inference process. For each query q, we begin by applying the "Unified Retrieval" module to extract key insights, denoted as D_{ins}. These insights serve as a sub-corpus for performing active retrieval during the MCTS process.

As outlined in Algorithm 1, distinct retrievers are utilized to handle text-only and multimodal corpora separately. The top-K knowledge retrieved from both routes is then combined to form $D_{\text{top-K}}$. This set of documents undergoes further refinement through the "Knowledge Concept Filter," yielding the final corpus of key insights, D_{ins}.

Once the key insights are obtained, the AR-MCTS inference algorithm, detailed in Algorithm 2, is executed. During each expansion step t, given a beam size B, the top-B most relevant insights are retrieved from D_{ins}. Each retrieved document is paired with the previous state s_{t-1} and fed into independent paths, where the multimodal large language model (MLLM), $\mathcal{M}$, generates the next state. The Process Reward Model (PRM), π_θ, then evaluates the candidate states $s_{(t,1)}, \ldots, s_{(t,B)}$ and assigns PRM scores to each. The state with the highest PRM score is appended to the selected path $\mathcal{P}$. This process continues until a terminal state is reached, resulting in the final reasoning trajectory and the corresponding answer.

B More Details about Experimental Setup

B.1 Benchmarks and Datasets

Here are the details of the benchmarks/datasets we used in our hybrid-modal retrieved corpus and experiments. The statistics of the datasets are recorded in Table 3.

- **We-Math** (Qiao et al., 2024) is a benchmark based on textbook knowledge units, focusing on decomposing complex problems into sub-problems using fundamental concepts. It mirrors how students learn progressively and is organized hierarchically, following textbook content to maintain independent knowledge units while establishing logical connections between levels. It uses diverse evaluation metrics to comprehensively assess models' ability of solving multimodal mathematical problems step by step.

- **MathVista** (Lu et al., 2024b) is a mathematical visual benchmark consisting of 6,141 examples. These examples are divided into two subsets: *testmini* (1,000 examples), for which answers are provided, and *test* (5,141 examples), for which answers are not publicly available. We use this dataset as a benchmark to evaluate visual understanding and compositional reasoning abilities. Additionally, we employ LLaVA-OneVision-70B to generate answers for the *test* split, creating an in-domain corpus that can be used for retrieving answers in the *testmini* set.

- **MathVision** (Wang et al., 2024b) is a carefully curated dataset consisting of 3,040 high-quality mathematical problems, each accompanied by a visual context derived from real mathematics competitions. The collection covers 16 distinct mathematical domains and is categorized across five levels of difficulty. We use this dataset as part of math reasoning knowledge base.

- **MathVerse** (Zhang et al., 2024c) is a comprehensive and specialized visual mathematics benchmark designed to evaluate the multimodal mathematical reasoning abilities of

Table 3: The statistics of General and Math-Specific Reasoning Knowledge.

Dataset	Count	Percentage
General Knowledge		
Wikipedia(zh-CN)	4.7B	23.9%
Wikipedia(en-US)	15B	73.6%
COIG	178K	0.1%
Mathematics-Specific Knowledge		
Text-only Datasets		
GSM8K	8,792	24.6%
MATH	12,500	36.2%
Multimodal Datasets		
MathVista	6,141	17.8%
MathVerse	2,612	7.6%
MathVision	3,040	8.8%
We-Math	1,740	5.0%

Algorithm 1 Unified Retrieval

Require: Query q, hybrid-modal retrieval corpus D_{H}, top-K, cross-modal retriever R_c, text-to-text retriever R_t

Ensure: Top-K retrieved hybrid-modal samples $D_{\text{top-K}}$

for all $d_i \in D_{\text{H}}$ **do** ▷ Text-Only Retrieval
  Query embedding $E_q \leftarrow R_t(q)$
  Document embedding $E_{d_i} \leftarrow R_t(d_i)$
  Retrieved documents $D_{\text{text}} = \text{argtop}_K^{i=1,\ldots,N} \left[E_{d_i}^{\top} \cdot E_q \right]$
end for
for all image-text pair $(x, t) \in D_{\text{H}}$ **do** ▷ Cross-modal Retrieval
  Image embedding $E_I(x) \leftarrow R_c(x)$
  Text embedding $E_T(t) \leftarrow R_c(t)$
  if $t \neq \emptyset \wedge x \neq \emptyset$ **then**
    $E_x(x, t) \leftarrow \frac{E_I(x)+E_T(t)}{2}$
  else if $t \neq \emptyset \wedge x = \emptyset$ **then**
    $E_x(x, t) \leftarrow E_T(t)$
  end if
  Mixed vector $E_x(Q^m) \leftarrow \frac{E_I(x)+E_T(t)}{2}$
  Retrieved documents $D_{\text{cross}} \leftarrow \text{argtop}_K^{j=1,\ldots,N} \left[E_x(Q^m)^{\top} \cdot E_x(x_j, t_j) \right]$
end for
$D_{\text{top-K}} \leftarrow \{D_{\text{q}} \cup D_{\text{cross}}\}$

Require: Knowledge concept label L_{kc}, original retrieval threshold T_r, knowledge concept consistency threshold T_{kc}

Ensure: Key insights D_{ins} for query q

Key insights $D_{\text{ins}} = \{r \in D_{\text{H}} \mid \text{Sim}(r, q) \geq T_r \ \& \ \text{Sim}(r, L_{\text{kc}}) \geq T_{\text{kc}}\}$ ▷ Knowledge Concept Filtering

Algorithm 2 Inference with AR-MCTS

Require: Beam Size B, question q, Process Reward Model π_θ, max depth T, MLLM $\mathcal{M}$, multimodal retriever R

Ensure: Selected path (thought process and answer) $\mathcal{P}$

$\mathcal{P} = [s_0]$, $t = 0$ ▷ Initialize Selected Path
while $t < T \wedge$ non-terminal path in $\mathcal{P}$ **do**
  Retrieved insights D_{ins}
  $D_{\text{top_B}} \leftarrow R(\mathcal{P}, D_{\text{ins}})$
  for all $d_i \in D_{\text{top_B}}$ **do**
    $s_{(t,i)} \leftarrow \mathcal{M}(\mathcal{P}, d_i)$
    PRM score $\text{score}(s_{t,i}) = \pi_\theta(s_{t,i})$
  end for
  $j \leftarrow \text{index}(\text{argmax}(\text{score}))$
  Add $s_{(t,j)}$ to $\mathcal{P}$
  Increment $t \leftarrow t + 1$
end while

MLLMs. It comprises a dataset of 2,612 visual math problems, with 1,236 newly acquired from public question repositories and 1,376 sourced from existing benchmarks. Each problem has been transformed by human annotators into six distinct versions—text-dominant, text-lite, text-only, vision-intensive, vision-dominant, and vision-only—each offering different levels of multimodal information. In our study, we utilize the "vision-only" version as image data and the "text-only" version as textual data to construct a knowledge base. This dataset is employed solely for the purpose of knowledge base construction and not as a benchmark.

- **MATH** (Hendrycks et al., 2021) is a dataset comprising 12,500 challenging competition mathematics problems. Each problem includes a comprehensive step-by-step solution, which can be used to train models in generating answer derivations and explanations. The dataset features problems from various mathematics competitions, including the AMC 10, AMC 12, AIME, and others. We utilize this dataset as a text-only corpus for mathematical domain reasoning.

- **GSM8K** (Cobbe et al., 2021b) (Grade School Math 8K) is a dataset containing 8,500 high-quality, linguistically diverse grade school math word problems. This dataset was designed to support question-answering tasks for basic mathematical problems requiring multi-step reasoning. This dataset is also used as part of our text-only mathematics-specific reasoning corpus.

- **COIG** (Zhang et al., 2023) (Chinese Open Instruction Generalist) is a set of Chinese instruction datasets to advance the training and fine-tuning of Chinese LLMs. COIG includes five key corpora: a manually verified translated instruction corpus (66,858 entries), an exam-based Chain-of-Thought (CoT) instruction corpus derived from national exams (63,532 entries), a human value alignment corpus reflecting general and region-specific cultural values (34,471 entries), a counterfactual correction multi-round chat corpus addressing hallucinations and inconsistencies (13,653 dialogues), and a Leetcode instruction corpus supporting code-related tasks (11,737 entries). We utilize this dataset along with the Wikipedia corpus (English version and Chinese version) as our general reasoning knowledge base.

- **GAOKAO-MM** (Zong and Qiu, 2024) is a comprehensive Chinese multimodal benchmark designed based on the Chinese National College Entrance Examination (Gaokao). It encompasses eight academic subjects and includes twelve categories of images, such as diagrams, function graphs, maps, and photographs. The benchmark aims to evaluate models' abilities to understand and reason over diverse multimodal content, reflecting the complexity and breadth of knowledge. We construct the domain-specific knowledge base using questions and answers from the years 2010 to 2021, while employing the questions from 2022 and 2023 as the test set.

B.2 Baselines and Backbone Models

To assess the gains from our approach, we compare it against a number of baselines as follows.

- **Self-Consistency** (Wang et al., 2023) involves sampling multiple reasoning paths from a large language model. Since each path may lead to different final answers, Self-Consistency selects the most consistent answer as the final output by marginalizing these sampled paths. This method is based on the intuition that complex reasoning problems often have a unique correct answer that can be reached through various approaches.

- **Self-Correction** (Madaan et al., 2023) is an iterative refinement method that improves the output of large language models (LLMs) or Multimodal large language models (MLLMs) through self-feedback. The core idea is to mimic the human revision process in writing: first, a preliminary output is generated, feedback is provided on this output, and improvements are made iteratively based on the feedback. Notably, this process allows for iterative optimization.

- **ORM** (Cobbe et al., 2021a) samples data from the reasoning training set to obtain result-oriented annotations for each sampled path. These data are then used to train a verifier that assists the generator in identifying higher-quality reasoning paths during prediction. In

this paper, we use the same data as for training PRM, with the distinction that the annotations are made directly using ground truth results rather than through AR-MCTS for step-level annotations, training ORM to assess the quality of reasoning paths.

In addition, we provide a detailed introduction to the MLLMs used in our experiments and their corresponding language backbone models.

- **Qwen2-VL** (Wang et al., 2024e), developed by Alibaba Cloud, represents an advanced iteration of the Qwen-VL series. By employing the Naive Dynamic Resolution mechanism, it dynamically processes images with varying resolutions and aspect ratios. The model achieves state-of-the-art performance on several visual understanding benchmarks, including MATHVISTA, MathVision, and WE-MATH.

- **InternVL2** (Chen et al., 2024d) family comprises multimodal large language models designed for advanced multimodal understanding tasks, demonstrating performance competitive with proprietary systems. Built using a progressive alignment training strategy, InternVL2 supports multimodal inputs, generalizes across diverse downstream tasks, and spans models ranging in size from 1 billion to 108 billion parameters. The InternVL2-8B variant exhibits remarkable capabilities in complex reasoning and shows promise for mathematical problem-solving applications.

- **LLaVA-NeXT** (Liu et al.) is a large-scale multimodal language model optimized through a cost-effective, realistic visual instruction-tuning dataset. It emphasizes enhanced visual reasoning, optical character recognition (OCR), and visual conversation capabilities. LLaVA-NeXT demonstrates superior performance across various multimodal benchmarks, including MMMU and MATHVISTA.

- **LLaVA-OneVision** (Li et al., 2024a) is a family of large-scale multimodal large language models (MLLMs) designed to extend the performance boundaries of open MLLMs across diverse scenarios, including single-image, multi-image, and video applications. It processes text, images, interleaved image-text inputs, and videos, supporting resolutions of up to 2304×2304 pixels. LLaVA-OneVision is available in various sizes, ranging from 0.5 billion to 72 billion parameters, and facilitates robust task transfer learning across modalities. Notably, it demonstrates exceptional video understanding by leveraging task transfer capabilities developed from image-based training.

- **GPT-4o** (OpenAI, 2024), a proprietary large-scale multimodal model developed by OpenAI, processes vision, text, and audio inputs. Built on Transformer architecture, the model is pre-trained on next-token prediction tasks and refined through a post-training alignment process. GPT-4o exhibits state-of-the-art multimodal understanding, achieving outstanding results across a variety of complex multimodal tasks.

- **GPT-4V** (OpenAI, 2023b), also developed by OpenAI, is a highly capable multimodal system enabling users to process image inputs and interleaved image-text data with GPT-4 models. It achieves impressive human-level performance across a broad spectrum of tasks, including scene text understanding, abstract reasoning, and open-world question answering.

- **Qwen2** (Yang et al., 2024) is a series of large language models (LLMs) based on the Transformer architecture, trained on a high-quality, diverse dataset of over 7 trillion tokens using next-token prediction. Spanning parameter sizes from 0.5 billion to 72 billion, Qwen2 is designed to enhance mathematical and coding reasoning capabilities. It achieves performance competitive with proprietary models across benchmarks for reasoning, language understanding, and generation. The Qwen2 series includes both foundational models and instruction-tuned variants, fine-tuned on datasets for single-turn and multi-turn instruction following.

- **InternLM2.5** (Cai et al., 2024) is a series of LLMs optimized for superior mathematical reasoning. Based on the InternLM2.5 foundational models, the series includes chat models fine-tuned through supervised fine-tuning (SFT) and reinforcement learning from human

feedback (RLHF), enabling robust instruction-following capabilities in downstream tasks. Notably, InternLM2.5-Chat-1M supports a 1-million-token context, demonstrating exceptional performance on long-context benchmarks.

- **Llama3** (Dubey et al., 2024) is a family of LLMs built on a standard dense Transformer architecture, natively supporting multilinguality, coding, reasoning, and tool integration. Pretrained on a large-scale, meticulously curated dataset, it undergoes post-training through supervised fine-tuning (SFT), rejection sampling (RS), and direct preference optimization (DPO). The flagship model, LLaMA3-405B, represents a significant scale-up from its predecessor, LLaMA2, trained on 15.6 trillion text tokens. It delivers competitive performance with GPT-4 across diverse benchmarks, including GSM8k, MATH, and MMLU.

B.3 Implementation Details

For uni-modal retrieval, we utilize mcontriever-mscoco, a multilingual version of Contriever (Izacard et al., 2022) fine-tuned on the MSMARCO dataset, as the text encoder. For multimodal retrieval, we employ the frozen CLIP model (ViT-L/14@336px variant) (Radford et al., 2021) as the multimodal encoder for both texts and images. To ensure the diversity and relevance of multimodal retrieval results, we incorporate five types of similarity measures: text-to-text, text-to-image, image-to-image, image-to-text, and cross-modal retrieval (introduced in Section 4.2) . Given the extensive size of the knowledge base, we leverage the open-source indexing engine FAISS (Johnson et al., 2021) to efficiently index dense vectors and retrieve the Top-k knowledge pieces.

For the Curriculum Process Reward Modeling, in the "Step-Wise DPO Pre-Alignment" phase, the learning rate is set to 5e-7 with a cosine scheduler and a 0.1 warm-up ratio. We use DeepSpeed ZeRO Stage 3 (Rasley et al., 2020) and Flash-Attention 2 (Dao, 2023) for efficiency, with a global batch size of 64. Training utilizes a sigmoid loss function with a beta value of 0.3 and spans 2 epochs, with checkpoints every 500 steps. Mixed precision training with bf16 is employed, and the maximum context length is 4096 tokens.

In the "Point-Wise Fine-Tuning" phase, we perform full fine-tuning on our PRM with a learning rate of 7e-6, using a linear scheduler with 20 warm-up steps. All models are trained with DeepSpeed ZeRO Stage 3 and Flash-Attention 2. We use a global batch size of 128, a weight decay of 0.1, and train for 3 epochs, saving checkpoints every 200 steps. Mixed precision training with bf16 is used, and the maximum context length is set to 8192 tokens. We run all our experiments on 8 NVIDIA A800 GPUs.

In our experiments, For the MATHVISTA, we picked 6 categories from the original 12: ALL (overall accuracy), GPS (geometry problem solving), MWP (math word problems), ALG (algebraic reasoning), GEO (geometry reasoning), and STA (statistical reasoning). For WE-MATH, we selected 8 categories: S1 (one-step problems), S2 (two-step problems), S3 (three-step problems), AVG (strict overall average scores), IK (insufficient knowledge), IG (inadequate generalization), CM (complete mastery), and RM (rote memorization).

B.4 Detailed Processing about Retrieved Corpus

For Chinese evaluation on GAOKAO-MM, we utilize the COIG dataset and Chinese Wikipedia dump which contains over 2.6 million articles as the textual knowledge base. We first apply the tool WikiExtractor (Attardi, 2015) to extract clean texts from the Wikipedia dump and remove low-resource articles, which results in over 1.3 million articles. We then split each article into disjoint passages of 256 characters, resulting in 4.7B passages. To enrich our knowledge base with more relevant in-domain information, we split GAOKAO-MM questions from 2010 to 2021 as a multimodal knowledge base. Since the evaluation is conducted on a more up-to-date subset of GAOKAO-MM from 2022 to 2023, we can effectively mitigate the risk of data leakage.

For English evaluation on MATHVISTA and WE-MATH, we choose COIG, GSM8K, MATH, and the English Wikipedia dump as the textual knowledge base. Following the same pre-processing steps of the Chinese Wikipedia dump, we obtain over 15B passages from the English Wikipedia dump as the basic retrieval unit. We have opted to employ MathVerse and MathVision as the multimodal knowledge base for their relevance to mathematical problem-solving and comprehension. Following MRAG-COT (Liu et al., 2023a), we use responses from LLaVA-OneVision (Li et al., 2024a)

as pseudo-answer for the test set of MATHVISTA. Due to the absence of an appropriate high-quality multimodal reasoning retrieval source or a training set with answer annotation, we incorporate the testmini set of WE-MATH into the knowledge base of MATHVISTA and include the testmini set of MATHVISTA into the knowledge base of WE-MATH.

B.5 Details about Knowledge Concept Filtering

As stated in the main text, high-quality labels are available for test sets like MATHVISTA and WE-MATH. However, not all external retrieval libraries or evaluation datasets have fine-grained concept labels (e.g., Wikipedia). To ensure the scalability of concept filtering, we use the open-world Tagger *InsTag* (Lu et al., 2024a) for offline knowledge concept annotation and repeat the aforementioned process for consistency filtering.

Specifically, we select the TagLM-13b-v2.0 model [4]. For text-only data, we directly annotated using InsTag and concatenated all coarse and fine-grained labels. For multimodal data, we generated captions for the images using the corresponding evaluation MLLM backbone, referencing InternVL2 (Chen et al., 2023b) and Vista (Zhou et al., 2024). We design the following caption generation template: "This is an image of a reasoning question; can you provide a detailed description of the image content?" We then concatenated the captions with the text and further used InsTag for annotation. After obtaining fine-grained labels, we followed the process outlined in the "Knowledge Concept Filtering" section for consistency screening.

B.6 PRM Training Data Collection.

As highlighted by MathPUMA (Zhuang et al., 2024), the challenge arises because the three multimodal benchmarks we evaluate lack training sets. Following the collection described in §3.2, we utilize four multimodal and two text-only datasets for process annotation, excluding any sources currently under evaluation. We extract multimodal QA pairs and use our AR-MCTS algorithm to automatically generate and annotate detailed solution processes. Notably, the GAOKAO-MM dataset is entirely in Chinese, which complicates reliance on English data sources. To address this, we classify data from 2010 to 2021 for AR-MCTS annotation, while questions from 2022 to 2023 serve as the test set.

[4] https://huggingface.co/OFA-Sys/TagLM-13b-v2.0

Table 4: Mathematical evaluation on MATHVISTA *testmini* sets. We select 6 out of the original 12 mathematical categories in MATHVISTA: ALL (overall accuracy), GPS (geometry problem solving), MWP (math word problems), ALG (algebraic reasoning), GEO (geometry reasoning), and STA (statistical reasoning). In the results for each model, the best accuracy scores are highlighted in **bold**.

Model	Method	ALL	GPS	MWP	ALG	GEO	STA
GPT-4V	Zero-shot	53.7	59.6	53.8	59.8	58.2	58.5
	Self-Consist.	56.2	65.4	53.2	63.7	**63.2**	58.8
	Self-Correct.	50.4	56.3	50.2	55.9	56.1	57.4
	ORM	56.6	65.3	53.1	**65.2**	**63.2**	59.0
	AR-MCTS	**57.4**	**66.1**	**53.9**	64.8	**63.2**	**59.5**
LLaVA-NEXT	Zero-shot	22.5	22.3	13.4	24.4	24.7	22.3
	Self-Consist.	23.1	22.6	16.7	26.0	24.3	24.3
	Seld-Correct.	22.5	22.6	17.2	24.9	22.6	25.2
	ORM	24.4	22.6	**17.5**	27.9	24.3	29.9
	AR-MCTS	**25.6**	**23.0**	17.4	**28.1**	**28.6**	**31.5**

B.7 PRM Training Guideline

Our PRM leverages the corresponding text backbone for evaluating MLLMs and consistently uses Qwen2-7B for closed-source models. Due to the lack of step supervision in multimodal reasoning, we collect existing open-source text-only PRM datasets, such as AlphaMath (Chen et al., 2024a), Math-Shepherd (Wang et al., 2024c), and PRM800K (Lightman et al., 2024). We first follow previous text-only works (Wang et al., 2024c; Zhang et al., 2024a) and perform preliminary fine-tuning alignment on our PRM backbone. Using the pre-aligned LLM, we apply the annotated data D_{align} from AR-MCTS and follow section "Curriculum Process Reward Modeling" to finalize the PRM. Consequently, we do not perform targeted fine-tuning on any MLLMs with in-domain data; instead, we focus on optimizing the PRM, significantly reducing computational resource consumption.

C More Details about Experimental Results

C.1 Results on More MLLMs backbones

To further validate the scalability of AR-MCTS, we conduct generalization studies on the widely used open-source MLLM Llama3-Llava-Next-8B and the powerful closed-source MLLM GPT-4V using MATHVISTA. As shown in Table 4, AR-MCTS continues to achieve stable improvements

Table 5: The contamination analysis on hybrid-modal retrieval corpus.

Dataset	MATHVISTA	WE-MATH
Text-only Datasets		
COIG	0.1%	0.1%
WIKIPEDIA(EN-US)	0.6%	1.1%
GSM8K	4.5%	2.0%
MATH	4.5%	1.8%
Multimodal Datasets		
MATHVERSE	0.7%	2.9%
MATHVISION	0.3%	0.9%
WE-MATH	0.5%	-
MATHVISTA-*testmini*	-	4.2%

and aligns with the three core conclusions from our main experiments:

- 1. MLLMs struggle to self-correct reasoning errors.
- 2. PRM outperforms ORM in complex reasoning tasks.
- 3. AR-MCTS better unlocks the reasoning potential of weaker MLLMs.

This further confirms the scalability of our core experimental findings.

C.2 Contamination Analysis on Hybrid-modal Retrieval Corpus

To further ensure that our hybrid-modal retrieval corpus does not contain any data leakage examples from the test set, we perform a data contamination analysis. We employ commonly used n-gram contamination algorithms to assess the overlap between the Top-50 samples retrieved by the retriever from different data sources and various test sets. As shown in Table 5, we follow the AUTOIF (Dong et al., 2024a) and test the n-gram threshold of 13. The results show that all data sources exhibited an overlap of less than 5% with MATHVISTA and WE-MATH. This highlights that there is no overlap between our retrieval library and the test sets.

C.3 Ablations on Different Retrievers

To validate the effectiveness of our general retrieval component, we conduct ablation studies by replacing different text and multimodal retrievers. Specifically, we used the following:

- **Text Retrievers:** BM25 (Sparse), Contriever (Dense)

Table 6: The ablations of different text retrievers.

Model	ALL	GPS	MWP	ALG	GEO	STA
Qwen2-VL-7B	58.8	45.5	60.5	45.5	47.9	70.8
+ BM25	60.2	54.8	57.9	53.3	54.6	72.1
+ Contriever	59.9	53.9	58.5	53.3	54.1	72.4

Table 7: The ablations of different multimodal retrievers.

Model	S1	S2	S3
Qwen2-VL-7B	53.4	37.2	33.9
+ CLIP-ViT-L/14	54.9	38.7	34.5
+ Jina-CLIP-v1	54.4	36.9	34.1

- **Multimodal Retrievers:** CLIP-ViT-L/14, Jina-CLIP-v1 (Koukounas et al., 2024)

The experimental results are presented in the Table 6 and 7, where we concatenated the top two retrieval results for each sample. The results indicate that different retrievers provide varying degrees of enhancement for downstream reasoning tasks. This not only demonstrates that our general retrieval module is plug-and-play but also highlights the rationale behind our mixed retrieval library.

C.4 Comparison of Different Training Objectives for PRMs

Table 8: The comparison of different training objectives for PRMs.

Model	ALL	GPS	MWP	ALG	GEO	STA
PRM (Hard)	62.9	63.3	71.5	59.4	62.2	71.0
PRM (Soft)	64.1	63.9	72.6	60.9	63.6	72.4

In this section, we explore the relationship between the training method of PRM and its multimodal reasoning capabilities. Following the setup of Luo et al., we investigate the use of hard labels versus soft labels trained through a linear layer connected to a large model.

As shown in Table 8, we find that PRM trained with soft labels performs better than that trained with hard labels on MATHVISTA.

Limitations

Despite our best efforts to optimize the AR-MCTS process, there are still several limitations and areas for improvement.

- **Computational Cost Optimization:** Annotating processes with MCTS algorithms requires significant computational resources, leading to high resource consumption—a common challenge in reasoning verification. However, AR-MCTS costs are still substantially lower than manual annotation. As a plug-and-play framework, AR-MCTS focuses on optimizing reasoning without the need to train multimodal foundational models, which significantly reduces computational overhead. The emergence of efficient techniques like vLLM (Kwon et al., 2023) is also helping to address this issue.

- **Exploration of PRM for Multimodal Model Foundations**: AR-MCTS represents a pioneering effort in step-wise reasoning within the multimodal domain, utilizing foundational training of MLLMs to align the PRM process. An ideal scenario would involve training the PRM within these models to enhance interactions between image and text and provide supplemental information for process-level reasoning. However, the lack of annotated data and higher computational demands present significant challenges in this area, which remains largely unexplored and is a direction for our future work.

- **Deep Integration of Retrieval and Reasoning**: Research highlights knowledge gaps in stepwise reasoning (Liu et al., 2024). AR-MCTS introduces a dynamic retrieval strategy that effectively addresses this issue. We believe this area still holds great potential for exploration, particularly in dynamically supplementing missing knowledge based on feedback from multimodal large models, which will be a key focus of our future research.

In summary, achieving trustworthy step-wise reasoning still presents challenges that await resolution and exploration. We sincerely hope that our team and researchers in the MLLM field can collaborate to address these issues in the future.